

**WYDZIAŁ FIZYKI
I INFORMATYKI STOSOWANEJ**

Real World Face Recognition Attacks Testing, Evaluating and Comparison

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1 Introduction

1.1 Motivation

The rapid development and widespread adoption of Artificial Intelligence (AI) technologies have significantly increased the use of face recognition systems in everyday applications. Face recognition is now commonly deployed in areas such as smartphone authentication, border control, surveillance systems, banking, and access management. Modern deep learning models achieve very high recognition accuracy and are often considered reliable enough for security-critical environments.

At the same time, the growing popularity of AI-based biometric systems has raised important concerns regarding their robustness and security. Numerous studies have demonstrated that deep neural networks can be vulnerable to adversarial attacks, where carefully crafted perturbations may cause incorrect predictions while remaining difficult or even impossible for humans to notice. In the context of face recognition, such attacks may allow unauthorized users to evade identification systems or impersonate other individuals [4].

However, despite the increasing deployment of face recognition technologies, the number of works focusing specifically on the security of real-world face recognition systems remains relatively limited compared to the broader AI research landscape.

1.2 Project Goal

The goal of this project is to investigate, implement, and compare multiple types of attacks against modern face recognition systems. By evaluating different attack methodologies and analyzing their effectiveness, this work aims to contribute to a better understanding of the vulnerabilities present in contemporary face recognition models.

2 Methodology

2.1 Dataset

All experiments were conducted using the Labelled Faces in the Wild (LFW) dataset. LFW is one of the most widely used benchmarks for unconstrained face recognition and verification tasks. The dataset contains 13,233 face images of 5,749 individuals collected from the Internet, with 1,680 subjects represented by two or more images. The images were automatically detected and aligned using the Viola–Jones face detector and are provided in a standardized format suitable for face recognition research. In this project, the deep-funneled version of the dataset was used, as it provides improved facial alignment and is commonly employed in modern face verification benchmarks.

The dataset was obtained from Kaggle: [2]

2.2 Face Recognition Attack Benchmark

To evaluate the robustness of the face recognition system against adversarial perturbations, a dedicated benchmark pipeline was implemented. The benchmark operates on pairs of images from the LFW dataset and measures how different image-level and face-level attacks affect embedding distances and final classification outcomes.

The system can be executed in two modes:

- full evaluation over all available pairs,
- a reduced quick test using a subset of pairs (e.g. `-n-pairs 300`).

Each attack is applied to one image in a pair, after which embeddings are recomputed and compared using a fixed decision threshold.

2.2.1 Configuration via YAML

The benchmark is configurable using a YAML file, which defines the list of attacks and their corresponding parameters. This allows for systematic and reproducible experimentation without modifying the core evaluation code. Each attack can be independently parameterized, enabling controlled comparison between different perturbation strategies.

2.2.2 Evaluation Procedure

For each image pair, the following steps are performed:

1. Compute baseline embeddings and distance between original images.
2. Apply a selected adversarial attack to one image.
3. Recompute embeddings and distance after the attack.
4. Compare classification labels before and after the attack.

A prediction is considered *fooled* if the model’s decision changes after the attack (e.g. from “same person” to “different person”).

2.2.3 Metrics

The benchmark reports:

- average embedding distance before and after the attack,
- percentage of fooled predictions (cases where the predicted identity changes after applying the attack),
- number of successfully evaluated pairs,
- number of skipped pairs before and after attack,
- distribution of baseline predictions (number of *same-person* vs. *different-person* classifications before attack),
- total runtime per attack.

By *True fooled rate*, we refer to the output of the following formula:

$$\frac{(\text{evaluated pairs} \cdot \text{percentage of fooled predictions}) + \text{number of skipped pairs after attack}}{\text{number of all pairs} - \text{number of skipped pairs before attack}}$$

which quantifies the overall effectiveness of the attack while accounting for both successful adversarial outcomes and evaluation coverage. All results are stored in structured JSON files as well as human-readable summaries, allowing for further analysis and reproducibility of experiments.

2.3 Threshold Selection Method

The decision threshold for face recognition is estimated using an empirical approach based on the distribution of distances between facial embedding vectors.

First, a subset of image pairs is sampled from the LFW dataset and divided into two groups:

- pairs of images of the same person,
- pairs of images of different people.

For each pair, facial embeddings are extracted and a distance metric is computed in the embedding space.

To obtain a robust decision boundary, we analyze the empirical distributions of these distances:

- the 95th percentile of distances for same-person pairs, denoted as $P_{95}(\text{same})$,
- the 5th percentile of distances for different-person pairs, denoted as $P_5(\text{different})$.

The final threshold is defined as the midpoint between these two values:

$$\text{threshold} = \frac{P_{95}(\text{same}) + P_5(\text{different})}{2}$$

This method places the threshold in the overlap region between the two distributions, providing a balance between false acceptance and false rejection rates without requiring full ROC curve optimization. The obtained decision threshold for the face recognition system is:

$$\text{threshold} = 0.7233$$

3 Face Recognition Attack Types

This section is based on the taxonomy presented in [3].

3.1 Image-Level Grid-Based Occlusion

3.1.1 Methodology

Attack mechanism. The grid-based occlusion attack is designed to simulate structured partial obstruction of facial regions by introducing deterministic or stochastic grid patterns over the input image. The core idea is to disrupt the spatial continuity of facial features by masking selected pixel regions with black lines, which forces the embedding model to operate on incomplete or heavily distorted facial representations.

The attack operates directly in the image space and does not require access to model gradients, making it a purely image-level perturbation method. Depending on configuration, the occlusion pattern may consist of vertical, horizontal, and optionally diagonal lines.

Grid-based occlusion generation process. Given an input image $I \in \mathbb{R}^{H \times W \times C}$, the attack generates a grid structure controlled by three main parameters:

- ρ_{grids} – number of grid lines controlling occlusion density,
- $line_width$ – thickness of each occlusion line,
- $random_spacing$ – determines whether grid lines are evenly spaced or randomly sampled,
- $diag_lines$ – enables additional diagonal occlusion patterns.

For the standard (non-diagonal) variant, vertical and horizontal line positions are generated either uniformly using linear spacing or randomly sampled from the image dimensions. Each selected coordinate defines a line region, which is then expanded by $line_width$ pixels and overwritten with zero intensity values, effectively removing visual information in those regions.

For the diagonal variant, additional sets of lines are generated along both main diagonals. These are defined using offset-based line equations ($j = i + c$ and $j = c - i$), where c controls the line position. The same masking procedure is applied to these diagonal regions.

Implementation details. The attack is implemented in Python using NumPy for array manipulation and PIL for image I/O operations. The image is processed in HWC format and copied before modification to preserve the original input.

The algorithm proceeds as follows:

1. Load image and convert it to a NumPy array.
2. Generate grid coordinates based on selected configuration.
3. For each coordinate, compute a rectangular mask region defined by $line_width$.
4. Set pixel values in masked regions to zero (black occlusion).

5. Optionally extend the process to diagonal patterns.

The function is deterministic for fixed parameters, except when *random_spacing* is enabled, in which case grid positions are sampled without replacement from uniform distributions over image dimensions.

The implementation is lightweight and computationally efficient, relying solely on vectorized array operations without requiring deep learning frameworks or gradient computation.

3.1.2 Results

The results of the grid-based occlusion attack experiments are summarized in Table 1. Each configuration modifies structural properties of the occlusion grid, including grid density (ρ), line thickness, randomness of spacing, and the presence of diagonal lines.

ID	ρ	Line width	Random spacing	Diagonal lines	True fool rate
1	8	1	No	No	1.0%
2	16	1	No	No	20.7%
3	16	1	No	Yes	16.7%
4	16	1	Yes	No	11.3%
5	16	1	Yes	Yes	13.7%
6	16	2	No	No	91.7%
7	16	2	No	Yes	95.3%
8	16	2	Yes	No	49.3%
9	16	2	Yes	Yes	60.0%
10	32	1	No	No	98.0%

Table 1: Corrected grid occlusion attack results ($N = 300$, threshold = 0.7233).

3.1.3 Analysis of Parameter Influence

The conducted experiments demonstrate that the effectiveness of grid-based occlusion attacks is strongly dependent on the structural parameters of the occlusion pattern, particularly grid density (ρ), line thickness, and the overall complexity of the generated mask.

Grid density (ρ): The results indicate a non-linear relationship between grid density and attack success rate. For low-density configurations ($\rho = 8$), the attack has negligible impact on the recognition system, achieving only a 1.0% fool rate. However, increasing the density to $\rho = 16$ leads to a significant increase in vulnerability, with fool rates ranging from 11.3% to 20.7%, depending on additional parameters. The most extreme configuration ($\rho = 32$) causes near-complete failure of the system, reaching a 98.0% fool rate. This suggests that higher spatial frequency of occlusion disrupts the stability of facial feature extraction in embedding space.

Line thickness: Line width is the most influential parameter in the attack. Increasing the thickness from 1 to 2 pixels results in a dramatic increase in attack effectiveness. While thin-line configurations remain relatively moderate (1.0%–20.7%), thicker grids achieve substantially higher fool rates, ranging from 49.3% to 95.3%. This indicates that the amount of occluded facial area plays a critical role in degrading embedding quality, likely by removing discriminative facial regions.

Random spacing: Introducing randomness in grid spacing does not consistently increase attack effectiveness. In some configurations, random spacing slightly reduces the fool rate (e.g., ID 4 vs. ID 2), while in others it has minimal impact. This suggests that stochastic perturbation of grid structure does not significantly improve transferability or robustness of the attack, and its influence is secondary compared to density and line thickness.

Diagonal lines: The addition of diagonal lines produces only marginal changes in performance. In most cases, configurations with diagonal components show slightly higher or comparable fool rates, but without a consistent trend. This indicates that diagonal structure increases visual complexity but does not fundamentally alter the feature disruption mechanism.

Overall observation: The results clearly show that grid-based occlusion attacks exhibit threshold-like behavior. Below a certain level of structural density and occlusion coverage, the model remains relatively robust. However, once the occlusion reaches sufficient spatial coverage (high ρ and line width = 2), the face recognition system rapidly degrades, indicating a breakdown of embedding consistency.

3.1.4 Grid-based Attack Visualization Example

To better illustrate the effect of grid-based occlusion, Figure 1 presents an example comparison between an original face image and its corresponding grid-occluded version used in the attack pipeline (with parameters: 24, 1, no, no):



Figure 1: Example of grid-based occlusion attack applied to a face image.

This visualization highlights how grid masking disrupts the continuity of facial features, forcing the recognition model to rely on incomplete spatial patterns rather than coherent identity information.

3.1.5 Conclusion

Grid-based occlusion attacks demonstrate that face recognition systems are highly sensitive to structured spatial perturbations. In particular, increasing grid density and line thickness significantly amplifies attack effectiveness, leading to substantial increases in embedding distances and misclassification rates. The most effective configuration ($\rho = 32$) indicates that dense occlusion patterns can almost completely break the recognition system under evaluation conditions.

These results highlight the importance of robustness against spatially structured occlusions, especially in real-world scenarios where partial face obstruction is common.

3.2 MSB Noise (XMSB)

3.2.1 Methodology

- definition of MSB-based perturbation
Modern RAM has vulnerability to flip bits outside the permitted memory. It is caused by hardware construction and law of physics and there exists methods to flip specific bit in memory, although it is slow process. In most cases it is used to change parameters of model, but it's also possible to corrupt photo of person to make future classifications, correlated to that person not working. There are trials to corrupt photo by flipping only one significant bit, but in this paper there was used 50 bits to flip, due to lack of results for smaller amount. For human eye photo before and after attack looks the same.
- noise generation strategy
Bits to flip were chosen by implementing genetic algorithm, which looks for the coordinations of

bits to flip, which gives biggest cosinus distance between corrupted photo and other photos of the same person.

3.2.2 Evaluation

To create a benchmark, different methods were used compared to the rest of the work, due to the specific nature of the attack. For the research, one person with 4 different photos was chosen. The cosine distance between one of the photos and the rest was calculated (the distance was always calculated after extracting embeddings using a pre-existing model), both before and after the attacks.

- Accuracy of the algorithm for uncorrupted photos

For the chosen person, there were 4 photos. After embedding, the cosine distance was calculated. Given an assumed threshold of 0.73, every pair was classified as the same person.

ID	Cosine distance
1	0.29561
2	0.32069
3	0.33255

- Degradation of recognition performance

To achieve the best possible results in every iteration of the algorithm for each genome, the sum of cosine distances between the corrupted photo and the rest of them (including the original photo before corruption) was calculated. The graph of the rising sum is shown below.

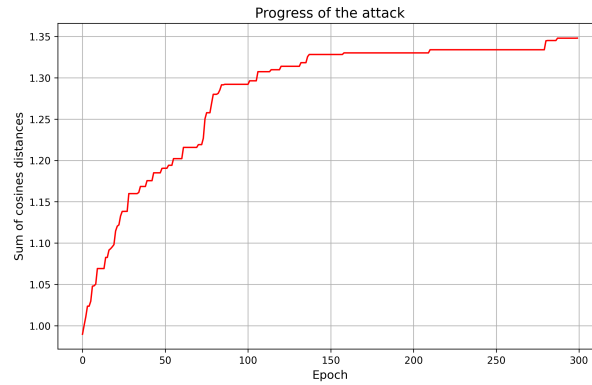


Figure 2: Relationship between the epoch of the algorithm and the progress of the attack

- Results and observations

For the best genome found by the algorithm, the results of the cosine distances between the corrupted photo and the rest of them can be found below:

ID	Cosine distance
1	0.41883
2	0.39203
3	0.46412

Even though the distance between the embeddings increased, it is nowhere near the 0.73 level, which is the threshold for the algorithm to decide that the persons are different. Giving the algorithm more time to work could slightly improve the results, although it is possible that there is a need to increase the number of flipped pixels.

3.3 Face-Level Distortion (Eye/Nose/Mouth Occlusion)

3.3.1 Methodology

Attack mechanism. Face-level occlusion attacks operate directly on semantically meaningful regions of the human face, such as the eyes, nose, and mouth. Unlike image-level perturbations that introduce artificial patterns (e.g., grids), these attacks simulate realistic partial occlusions that may occur in real-world scenarios, including accessories, shadows, or physical obstructions.

The core idea is to selectively mask biologically and semantically important facial regions in order to disrupt the feature extraction process of the face recognition model. Since modern embedding-based systems rely heavily on localized facial features, occluding these regions can significantly degrade recognition performance.

Occlusion generation process. Given an input image $I \in \mathbb{R}^{H \times W \times C}$, the attack applies rectangular occlusion masks centered on predefined facial regions: eyes, nose, or mouth. The size of the occluded region is controlled by two parameters:

- $padding_x$ – horizontal expansion of the occlusion region,
- $padding_y$ – vertical expansion of the occlusion region.

Each attack variant targets a specific facial region detected via landmark-based alignment. The occlusion is applied by overwriting the selected region with zero intensity values, effectively removing all visual information from that area.

Depending on the configuration, multiple levels of occlusion severity are tested by increasing padding parameters, resulting in progressively larger masked regions.

Implementation details. The attack is implemented using facial landmark detection followed by region-based masking in NumPy array representation. The pipeline consists of the following steps:

1. Detect facial landmarks and localize eyes, nose, or mouth region.
2. Compute bounding box around the selected region.
3. Expand the bounding box using $padding_x$ and $padding_y$.
4. Apply zero-valued mask to the selected pixels.
5. Preserve the remaining facial structure.

The method is deterministic given fixed landmark detection and padding parameters.

3.3.2 Results

The results of the face-level occlusion attacks are summarized in Table 2. Each configuration targets a specific facial region with varying occlusion intensity.

ID	Target region	Padding (x, y)	True fooled rate
1	Eyes	(15, 15)	4.00%
2	Eyes	(25, 20)	9.70%
3	Eyes	(40, 35)	54.31%
4	Nose	(20, 30)	7.30%
5	Nose	(30, 40)	39.65%
6	Nose	(45, 60)	98.33%
7	Mouth	(30, 30)	13.69%
8	Mouth	(50, 50)	69.97%
9	Mouth	(70, 70)	100.00%

Table 2: Face-level occlusion attack results (N = 300, threshold = 0.7233).

3.3.3 Analysis of Parameter Influence

The experiments demonstrate that face-level occlusion attacks are highly sensitive to both the targeted facial region and the extent of the occlusion.

Eyes region: Occlusion of the eye region shows a strong non-linear effect on model performance. Small padding values (15, 15) result in minimal impact (4.0% fool rate), while moderate occlusion increases the effect to 9.7%. However, larger occlusion (40, 35) leads to a sharp increase in attack success, reaching 54.31%.

Nose region: The nose region exhibits progressively increasing vulnerability as occlusion size grows. While small and medium configurations yield moderate fool rates (7.3% and 39.65%), large occlusion (45, 60) causes near-complete failure of the system (98.33%). This suggests that the central facial region plays a critical role in identity representation and is highly sensitive to distortion.

Mouth region: The mouth region shows a similar but slightly delayed degradation pattern. Initial occlusion has a limited effect (13.69%), but increasing the masked area significantly boosts attack effectiveness, culminating in a 69.97% fool rate for the medium configuration and a full 100.0% for the largest configuration. This indicates that while the mouth is less critical than the eyes or nose at small perturbation levels, extensive occlusion still strongly disrupts recognition.

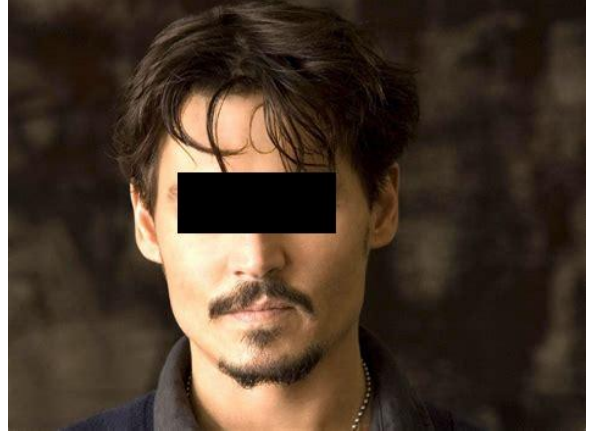
Overall observation: The results confirm that face recognition models rely on a distributed but non-uniform representation of facial features. The nose and eyes appear to contribute most strongly to identity encoding, while the mouth becomes increasingly influential as occlusion size grows. Once a sufficient portion of the central facial region is masked, the embedding space becomes unstable, leading to severe degradation and eventual complete failure of the recognition system in extreme cases.

3.3.4 Face-level Distortion Visualization Example

Figure 3 shows a illustration of landmarks position as well as face-level distortion effects.



(a) Landmarks



(b) Eyes occluded



(c) Mouth occluded



(d) Nose occluded

Figure 3: Examples of face-level distortion attacks & landmarks.

3.3.5 Conclusion

Face-level occlusion attacks demonstrate that semantic facial regions play a crucial role in embedding-based recognition systems. Unlike artificial grid-based perturbations, these attacks show that realistic occlusions can severely degrade performance, especially when central facial features such as the nose and eyes are heavily masked.

The results highlight the importance of robustness against partial face visibility and suggest that future recognition systems should incorporate stronger invariance to localized occlusions.

3.4 Feature-Level Attacks (FFM / FIM)

3.4.1 Methodology

Attack mechanism. Feature-level attacks operate by directly modifying the input image in order to change its representation in the embedding space. Unlike occlusion-based methods, which remove visible image regions, FFM and FIM generate small pixel-level perturbations constrained by an L_∞ budget ϵ . The goal is to move the attacked image away from its original embedding, increasing the embedding distance and forcing the recognition system to classify a pair differently.

The evaluated face recognition uses InsightFace for final verification. Since this model is used only for inference and does not expose gradients, the attack is optimized on a differentiable embedding model and then evaluated on InsightFace. In this project, the differentiable model is an InceptionResnetV1 network from `facenet-pytorch`, pretrained either on `vggface2` or `casia-webface`. This makes the attack a transfer-based feature-level attack.

Feature Fast Method (FFM). FFM is a single-step attack. First, the original image is converted to a tensor and passed through the model to obtain the reference embedding. Then the gradient of the similarity between the current embedding and the reference embedding is computed with respect to the input image. The adversarial image is updated once in the direction that decreases this similarity. The update is clipped to the selected perturbation budget, ensuring that no pixel changes by more than ϵ .

Feature Iterative Method (FIM). FIM extends FFM by applying several smaller gradient steps. Instead of using the full perturbation budget in a single update, it repeatedly modifies the image using step size α over a fixed number of iterations. After each step, the image is projected back into the allowed ϵ -bounded perturbation range and pixel values are clipped to the valid range $[0, 255]$.

Face-region masking. Most experiments restrict perturbations to the detected face region. The face bounding box is obtained using InsightFace and expanded by a configurable `face_margin`. Pixels outside this mask are not updated. This makes the perturbation more semantically focused and avoids modifying unrelated background regions. If no face is detected during mask construction, the implementation falls back to perturbing the full image.

Implementation details. The attack pipeline is implemented in PyTorch. Input images are resized to the model input size and preprocessed using like a FaceNet normalization. Embeddings are normalized before computing cosine similarity. The final adversarial image is rounded, clipped, and converted back to an unsigned 8-bit image before being passed to the benchmark evaluator.

The implementation supports the following parameters:

- ϵ – maximum allowed pixel change under the L_∞ constraint,
- *steps* – number of iterative updates used by FIM,
- α – step size used in each FIM iteration,
- *pretrained* – model used to expose gradients,
- *face_only* – whether perturbation is restricted to the detected face region,
- *face_margin* – expansion factor for the face-region mask.

3.4.2 Results

The feature-level attack experiments were evaluated on 300 image pairs using the threshold 0.7233. In the table, this value was computed from exact fooled-pair counts extracted from the JSON benchmark outputs. Therefore, skipped pairs after the attack are treated as successful attack outcomes because the perturbed image prevented the system from completing normal recognition.

ID	Attack	ϵ	Steps	α	Surrogate	Face mask	Avg. before	Avg. after	Skipped after	True fooled rate
1	FFM	8	1	8.0	vggface2	Yes, margin 0.25	0.3668	0.4386	0	1.33%
2	FFM	16	1	16.0	vggface2	Yes, margin 0.25	0.3664	0.5185	1	10.33%
3	FFM	24	1	24.0	vggface2	Yes, margin 0.25	0.3646	0.5751	3	30.00%
4	FIM	8	10	0.8	vggface2	Yes, margin 0.25	0.3668	0.4527	0	0.67%
5	FIM	16	10	1.6	vggface2	Yes, margin 0.25	0.3668	0.5498	0	6.00%
6	FIM	32	10	3.2	vggface2	Yes, margin 0.25	0.3668	0.6574	0	26.67%
7	FIM	16	1	16.0	vggface2	Yes, margin 0.25	0.3664	0.5185	1	10.33%
8	FIM	16	20	0.8	vggface2	Yes, margin 0.25	0.3668	0.5446	0	5.00%
9	FIM	16	10	2.0	vggface2	Yes, margin 0.25	0.3668	0.5721	0	8.33%
10	FIM	16	10	1.6	casia-webface	Yes, margin 0.25	0.3668	0.4611	0	1.33%
11	FIM	16	10	1.6	vggface2	No	0.3668	0.5446	0	5.00%
12	FIM	16	10	1.6	vggface2	Yes, margin 0.00	0.3668	0.5571	0	7.33%
13	FIM	16	10	1.6	vggface2	Yes, margin 0.50	0.3668	0.5436	0	5.00%

Table 3: Feature-level attack results for FFM and FIM (N = 300, threshold = 0.7233). True fooled rate is calculated with skipped-after pairs included according to the benchmark formula.

3.4.3 Analysis of Parameter Influence

Epsilon ϵ is the most important factor for FFM. Increasing ϵ from 8 to 16 raises the True fooled rate from 1.33% to 10.33%, and increasing it further to 24 raises it to 30.00%. This shows that the single-step feature-level attack transfers increasingly well to InsightFace when stronger pixel-level changes are allowed. The same tendency is visible for FIM: $\epsilon = 8$ gives only 0.67%, while $\epsilon = 32$ reaches 26.67%.

FFM compared with FIM: For the same $\epsilon = 16$ perturbation budget, the single-step configuration achieves 10.33%, while the standard iterative configuration with 10 steps and $\alpha = 1.6$ reaches 6.00%. This indicates that, in this setup, more iterations do not automatically improve transferability to the final InsightFace model. The one-step update may produce a stronger transferable direction, while the iterative process can overfit more closely to the surrogate embedding model.

Number of steps and step size: At $\epsilon = 16$, changing the FIM schedule noticeably affects performance. The one-step variant with $\alpha = 16$ reaches 10.33%, while 20 smaller steps with $\alpha = 0.8$ reach 5.00%. Increasing the 10-step step size from $\alpha = 1.6$ to $\alpha = 2.0$ improves the True fooled rate from 6.00% to 8.33%. This suggests that stronger individual updates were more effective than very fine iterative optimization for this transfer setting.

Proxy model choice: The surrogate model has a strong influence on transferability. With otherwise identical parameters ($\epsilon = 16$, 10 steps, $\alpha = 1.6$, face-only mask), the **vggface2** achieves 6.00%, while **casia-webface** achieves only 1.33%. This suggests that the gradients from the **vggface2** are better aligned with the embedding behavior of the evaluated InsightFace system.

Face mask and margin: Restricting perturbations to the detected face region does not always improve results. The full-image variant reaches 5.00%, while the default face-only margin 0.25 configuration reaches 6.00%. A tighter face mask with margin 0.00 performs better at 7.33%, whereas a larger margin 0.50 falls back to 5.00%. This indicates that perturbing the most identity-relevant facial region is more useful than spreading the same perturbation budget over a larger area.

Embedding distance: The attack generally increases average embedding distance after perturbation. For example, FFM with $\epsilon = 24$ increases the average distance from 0.3646 to 0.5751, and FIM with $\epsilon = 32$ increases it from 0.3668 to 0.6574. However, higher average distance does not always translate perfectly into a higher fooling rate, because only pairs crossing the fixed decision threshold change the final classification.

3.4.4 Feature-Level Attack Visualization Example

Figure 4 and 5 provides a place for visual comparison of the original image, the adversarially perturbed image, and an amplified perturbation map. The perturbation is difficult to interpret directly in the normal image scale, so the difference image is multiplied for visualization purposes.

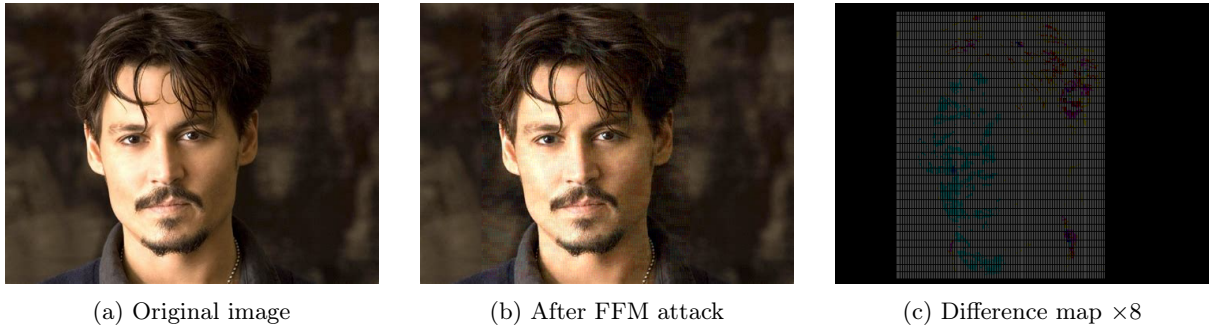


Figure 4: Example of a feature-level perturbation generated by FFM ($\epsilon = 8$).

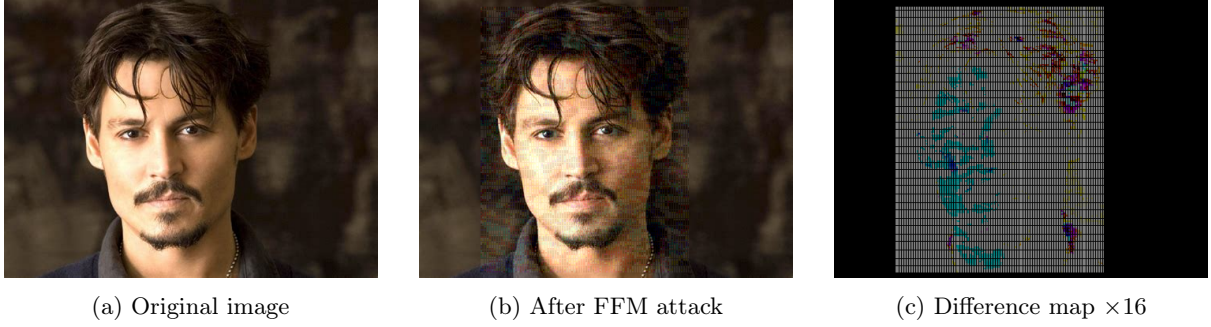


Figure 5: Example of a feature-level perturbation generated by FFM ($\epsilon = 16$).

3.4.5 Conclusion

Feature-level attacks demonstrate that the evaluated face recognition system is vulnerable not only to visible occlusions, but also to gradient-based perturbations optimized in embedding space. The strongest available FFM configuration reaches a True fooled rate of 30.00%, while the strongest available FIM configuration reaches 26.67%.

The experiments also show that transferability depends strongly on the epsilon, model used to gradient, and update schedule. In particular, single-step or larger-step attacks were more effective than very fine iterative optimization under the tested settings.

4 Results Analysis

4.1 Comparison of Attack Effectiveness

The conducted experiments reveal substantial differences in the effectiveness of the evaluated attack categories. Face-level, image-level, and feature-level attacks affect the face recognition system through fundamentally different mechanisms, resulting in varying levels of performance degradation.

Among all evaluated approaches, **Face-Level Occlusion** achieved the highest attack success rates. Large occlusions targeting the nose and mouth regions resulted in True fooled rates of 98.33% and 100.00%, respectively. These results indicate that the evaluated face recognition system relies heavily on central facial features and struggles to maintain embedding consistency when these regions are removed.

Grid-Based Occlusion also proved highly effective under sufficiently aggressive configurations. While sparse grids had limited impact, increasing grid density and line thickness caused a rapid increase in attack effectiveness. The strongest tested configuration ($\rho = 32$) achieved a True fooled rate of 98.0%, demonstrating that large-scale structured image corruption can almost completely disrupt recognition.

In contrast, **Feature-Level Attacks** (FFM and FIM) achieved noticeably lower success rates. The best-performing FFM configuration reached a True fooled rate of 30.0%, while the strongest FIM configuration achieved 26.67%. Although these attacks significantly increased embedding distances, they rarely pushed enough image pairs across the decision threshold to cause large-scale classification failures.

Finally, the **MSB Noise Attack** produced only limited degradation of recognition performance. Despite successfully increasing cosine distances between embeddings, the attack was unable to push distances beyond the decision threshold required for misclassification. Under the tested settings, no complete recognition failures were observed.

Overall, the results suggest that the evaluated face recognition system is considerably more vulnerable to attacks that remove or occlude meaningful facial information than to subtle pixel-level perturbations.

5 Conclusion

5.1 Final Remarks

This project investigated the robustness of a modern face recognition system against several categories of adversarial attacks. A dedicated benchmark framework was developed to evaluate attack effectiveness on

the LFW dataset using embedding-distance-based verification. The framework enabled systematic comparison of image-level, face-level, feature-level, and bit-flip-based attack methodologies under a common evaluation protocol.

Taken together, the results confirm that face recognition systems are not equally vulnerable to all attack classes. Attacks that directly disrupt identity-related visual information appear significantly more effective than subtle perturbations designed to manipulate embedding space.

5.2 Future Work

Several directions for future research can be identified based on the findings of this project.

First, the evaluation could be extended to larger datasets and additional face recognition models in order to assess the generality of the observed vulnerabilities. Comparing multiple commercial and open-source systems would provide a broader perspective on attack transferability and robustness.

Second, more advanced adversarial techniques could be investigated, including targeted impersonation attacks or better black-box optimization methods. Such attacks may better represent realistic threat scenarios encountered in deployed systems.

Third, defensive mechanisms could be explored. Potential approaches include adversarial training, input preprocessing, embedding-space regularization, occlusion-aware recognition architectures, and ensemble-based verification methods. Evaluating the effectiveness of these defenses against the attacks considered in this work would provide valuable insights into practical mitigation strategies.

Finally, future studies could focus on physical realizability by investigating attacks that can be executed outside the digital domain, such as adversarial eyeglasses, printed patterns, or face coverings. Understanding the gap between digital and real-world attack effectiveness remains an important challenge for the secure deployment of face recognition technology.

6 Github repository

Github repository is available here: [1]

References

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